

PMSCS 694

Information Security

Crypto: Part III

Lecture 5

Md. Rafsan Jani
Associate Professor
Department of Computer Science and Engineering
Jahangirnagar University

Block Ciphers



(Iterated) Block Cipher

- ❑ Plaintext and ciphertext consist of fixed-sized blocks
- ❑ Ciphertext obtained from plaintext by iterating a **round function**
- ❑ In most designs, the ciphertext is obtained from the plaintext by iterating a function F over some number of rounds.
 - The function F , which depends on the output of the previous round and the key K , is known as the round function, not because of its shape, but because it is applied over multiple rounds.
- ❑ In most designs, the ciphertext is obtained from the plaintext by iterating a function F over some number of rounds.
- ❑ The function F , which depends on the output of the previous round and the key K , is known as the round function, not because of its shape, but because it is applied over multiple rounds.
- ❑ Input to round function consists of **key** and **output** of previous round
- ❑ Usually implemented in software

Feistel Cipher: Encryption

- ❑ **Feistel cipher** is a type of block cipher
 - **Not** a specific block cipher

- ❑ Split plaintext block into left and right halves: $P = (L_0, R_0)$

- ❑ For each round $i = 1, 2, \dots, n$, compute

$$L_i = R_{i-1}$$

$$R_i = L_{i-1} \oplus F(R_{i-1}, K_i)$$

where F is **round function** and K_i is **subkey**

- ❑ Ciphertext: $C = (L_n, R_n)$

Feistel Cipher: Decryption

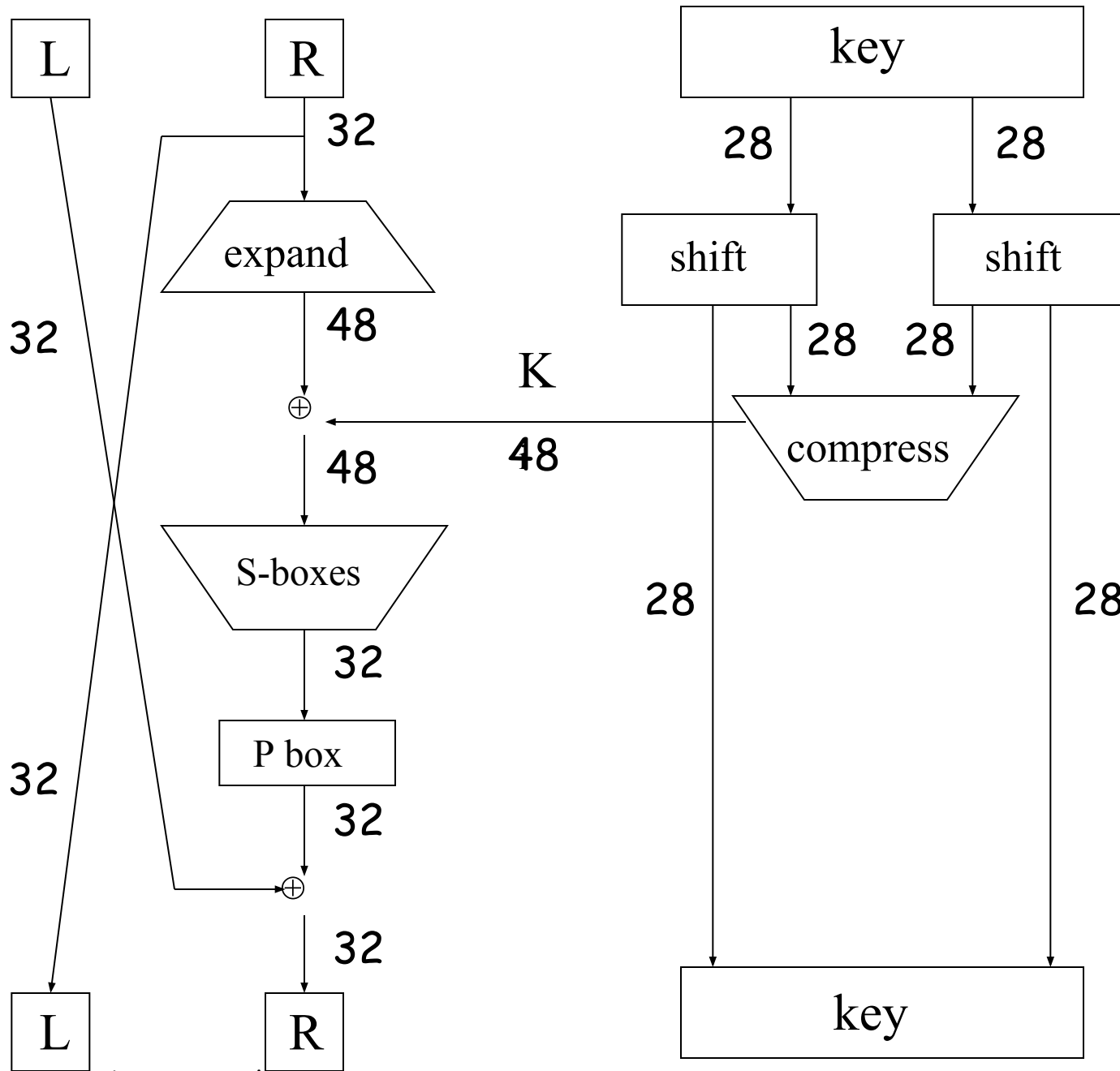
- Start with ciphertext $C = (L_n, R_n)$
- For each round $i = n, n-1, \dots, 1$, compute
$$R_{i-1} = L_i$$
$$L_{i-1} = R_i \oplus F(R_{i-1}, K_i)$$
where F is round function and K_i is subkey
- Plaintext: $P = (L_0, R_0)$
- Decryption works for any function F
 - But only *secure* for certain functions F

Data Encryption Standard

- ❑ **DES** developed in 1970's
- ❑ Based on IBM's Lucifer cipher
- ❑ DES was U.S. government standard
- ❑ Development of DES was controversial
 - NSA was secretly involved
 - Design process was secret
 - Key length reduced from 128 to 56 bits
 - Subtle changes to Lucifer algorithm

DES Numerology

- ❑ DES is a Feistel cipher with...
 - 64 bit block length
 - 56 bit key length
 - 16 rounds
 - 48 bits of key used each round (subkey)
- ❑ Round function is simple (for block cipher)
- ❑ Security depends heavily on "S-boxes"
 - Each S-box maps 6 bits to 4 bits



One
Round
of
DES

DES Expansion "Permutation"

□ Input 32 bits

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31

□ Output 48 bits

31	0	1	2	3	4	3	4	5	6	7	8
7	8	9	10	11	12	11	12	13	14	15	16
15	16	17	18	19	20	19	20	21	22	23	24
23	24	25	26	27	28	27	28	29	30	31	0

DES S-box

- ❑ 8 “substitution boxes” or S-boxes
- ❑ Each S-box maps 6 bits to 4 bits
- ❑ Here is S-box number 1

input bits (0,5)

↓

input bits (1,2,3,4)

		0000	0001	0010	0011	0100	0101	0110	0111	1000	1001	1010	1011	1100	1101	1110	1111

00		1110	0100	1101	0001	0010	1111	1011	1000	0011	1010	0110	1100	0101	1001	0000	0111
01		0000	1111	0111	0100	1110	0010	1101	0001	1010	0110	1100	1011	1001	0101	0011	1000
10		0100	0001	1110	1000	1101	0110	0010	1011	1111	1100	1001	0111	0011	1010	0101	0000
11		1111	1100	1000	0010	0100	1001	0001	0111	0101	1011	0011	1110	1010	0000	0110	1101

DES P-box

□ Input 32 bits

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31

□ Output 32 bits

15	6	19	20	28	11	27	16	0	14	22	25	4	17	30	9
1	7	23	13	31	26	2	8	18	12	29	5	21	10	3	24

DES Subkey

- ❑ 56 bit DES key, numbered 0,1,2,...,55
- ❑ Left half key bits, LK

49	42	35	28	21	14	7
0	50	43	36	29	22	15
8	1	51	44	37	30	23
16	9	2	52	45	38	31

- ❑ Right half key bits, RK

55	48	41	34	27	20	13
6	54	47	40	33	26	19
12	5	53	46	39	32	25
18	11	4	24	17	10	3

DES Subkey

- For rounds $i=1, 2, \dots, 16$
 - Let $LK = (LK \text{ circular shift left by } r_i)$
 - Let $RK = (RK \text{ circular shift left by } r_i)$
 - Left half of subkey K_i is of LK bits

13 16 10 23 0 4 2 27 14 5 20 9
22 18 11 3 25 7 15 6 26 19 12 1

- Right half of subkey K_i is RK bits

12 23 2 8 18 26 1 11 22 16 4 19
15 20 10 27 5 24 17 13 21 7 0 3

DES Subkey

- For rounds 1, 2, 9 and 16 the shift r_i is 1, and in all other rounds r_i is 2
- Bits 8,17,21,24 of LK omitted each round
- Bits 6,9,14,25 of RK omitted each round
- **Compression permutation** yields 48 bit subkey K_i from 56 bits of LK and RK
- **Key schedule** generates subkey

DES Last Word (Almost)

- ❑ An initial permutation before round 1
- ❑ Halves are swapped after last round
- ❑ A final permutation (inverse of initial perm) applied to (R_{16}, L_{16})
- ❑ None of this serves any security purpose

Security of DES

- ❑ Security depends heavily on S-boxes
 - Everything else in DES is linear
- ❑ 40+ years of analysis revealed no back door
- ❑ Attacks? Essentially exhaustive key search
- ❑ **Inescapable conclusions**
 - Designers of DES knew what they were doing
 - Designers of DES were way ahead of their time (at least wrt certain cryptanalytic techniques)

Block Cipher Notation

- ❑ P = plaintext block
- ❑ C = ciphertext block
- ❑ Encrypt P with key K to get ciphertext C
 - $C = E(P, K)$
- ❑ Decrypt C with key K to get plaintext P
 - $P = D(C, K)$
- ❑ **Note:** $P = D(E(P, K), K)$ and $C = E(D(C, K), K)$
 - But $P \neq D(E(P, K_1), K_2)$ and $C \neq E(D(C, K_1), K_2)$ when $K_1 \neq K_2$

Triple DES

- ❑ 56 bit DES key is too small
 - Exhaustive key search is feasible
- ❑ But DES was everywhere, so what to do?
- ❑ **Triple DES** or **3DES** (112 bit key)
 - $C = E(D(E(P, K_1), K_2), K_1)$
 - $P = D(E(D(C, K_1), K_2), K_1)$
- ❑ **Why Encrypt-Decrypt-Encrypt with 2 keys?**
 - Backward compatible: $E(D(E(P, K), K), K) = E(P, K)$
 - And 112 is a lot of bits

3DES

- ❑ Why not $C = E(E(P, K), K)$ instead?
 - Trick question —still just 56 bit key
- ❑ Why not $C = E(E(P, K_1), K_2)$ instead?
- ❑ A (semi-practical) **known plaintext** attack
 - Pre-compute table of $E(P, K_1)$ for every possible key K_1 (resulting table has 2^{56} entries)
 - Then for each possible K_2 compute $D(C, K_2)$ until a match in table is found
 - When match is found, have $E(P, K_1) = D(C, K_2)$
 - Result gives us keys: $C = E(E(P, K_1), K_2)$

Sample Questions

- ❑ Describe the structure and operation of the DES algorithm, including its key size and block size.
- ❑ What role do S-boxes play in block ciphers like DES?
- ❑ Why DES Encrypt-Decrypt-Encrypt with 2 keys?

Sample Questions

- In DES cipher.
 - a. How many bits in each plaintext block?
 - b. How many bits in each ciphertext block?
 - c. How many bits in the key?
 - d. How many bits in each subkey?
 - e. How many rounds?
 - f. How many S-boxes?
 - g. An S-box requires how many bits of input?
 - h. An S-box generates how many bits of output?